4.2 ENGINEERING MATERIALS

If we would like to fabricate an engineering part, like a hacksaw blade, we must go in search of material, like high carbon steel high speed steel which possess desirable properties as will permit the blade to perform its function successfully while in use. That is, any tool or cutting materials , should have high strength , high toughness, high hardness and high corrosion resistance.

Tungsten carbide, vanadium carbide, molybdenum steels are used as tool or cutting materials advanced in technology depend increasingly on the development of better materials of corrosion and resistance to heat.

4.3 PROPERTIES OF MATERIALS

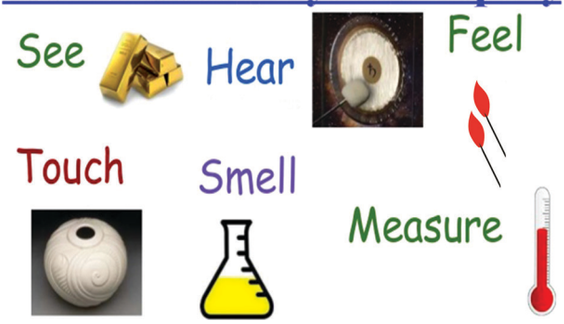
The practical application of engineering materials in manufacturing a component, we must have a thorough knowledge of their particular properties under a wide range of conditions. However, the range of properties found in different classes of materials in very large. Some of the very useful properties are,

1. Physical properties
2. Chemical properties
3. Electrical properties
4. Mechanical properties
5. Thermal properties

Physical Properties

Physical properties can be observed or measured without changing the composition of matter.To finalize the material for an engineering product or application, we should have the knowledge of physical properties of materials like the Density, Elasticity, Porosity, Thermal conductivity and Latent heat.

To know about Physical Property



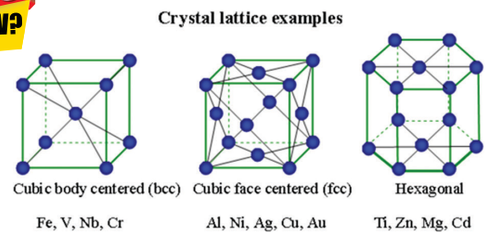
Chemical Properties

Being an Engineer, It is very important to have the knowledge of chemical properties of engineering materials. Because most of the engineering materials contact with other materials and chemically react to each other. Due to this chemical reaction they may suffers from chemical deterioration.

Some of the chemical properties of engineering materials are listed below

1. Chemical Composition
2. Atomic Bonding
3. Corrosion Resistance

DO YOU KNOW?



Crystal Structure - In crystallography, crystal structure is a description of the ordered arrangement of atoms, ions or molecules in a crystalline material.

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